

TASK 4

EXPLANATION

Here ESP32 acts as a Wi-Fi IoT client that streams sensor data to the Blynk IoT cloud using a persistent network session. The path is:

LDR → ESP32 → WiFi → Internet → Blynk Cloud → Mobile App/

Web- Console

Data Transmission Flow as:

1) Sensor acquisition (edge)

- `analogRead(GPIO36)` samples the LDR.
- ESP32 ADC converts analog voltage → digital value (0–4095).

2) Wi-Fi association (network layer)

- `WiFi.begin(ssid, password):`
 - ESP32 joins router (802.11).
 - Gets an IP via DHCP.
- At this point it can reach the internet (IP routing/NAT handled by router).

3) Session with Blynk server (transport/application)

- `Blynk.begin(AUTH, ssid, pass)` does two things:
 1. Connects Wi-Fi (if not already).
 2. Opens a persistent TCP socket to the Blynk cloud (default port typically 80/443/8080 depending on configuration).
- Authentication:
 1. The Auth Token uniquely identifies your device.
 2. Sent during session initialization to bind this device to your project.

4) Data packaging (application payload)

- In your timer function:

```
int ldrValue = analogRead(LDR_PIN);  
Blynk.virtualWrite(V0, ldrValue);
```

Here:

- V0 is a virtual pin (logical channel).

- IdrValue is serialized by the Blynk library into a compact protocol frame.

5) Transmission over TCP/IP

- The Blynk library pushes the frame through the open socket:
 - Application layer: Blynk protocol message
 - Transport layer: TCP (reliable, ordered delivery)
 - Network layer: IP packets
 - Link layer: Wi-Fi frames
- Because TCP is used:
 - Lost packets are retransmitted
 - Data arrives in order (important for real-time dashboards)

6) Cloud ingestion (Blynk server)

- Blynk cloud receives the message:
 - Verifies Auth Token
 - Maps V0 → your project's datastream
 - Stores latest value (and optionally history)

7) Visualization (mobile app/web console)

- Your phone app/ web console subscribes to the same datastream:
 - Gauge widget reads V0
 - LED indicator(status) reads V2
 - LED switch(control) via read V2
 - Speaker alert reads V2
- Updates are near real-time because the device keeps a live connection.

Reverse path (control from cloud to device)

Equally important—control signals flow back:

- You toggle switch (V1) in app/web
- Blynk server sends command → ESP32
- Callback executes:

```
BLYNK_WRITE(V1)
{state = param.asInt();
  digitalWrite(LED_PIN, state);
}
```

- LED changes state instantly

❖ **This makes it a bidirectional IoT system, not just monitoring.**